

ITA0303 - Mobile Computing

C01 - ASSESSMENT

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1. * The spectrum can support 900 full-duplex voice channels.

calculation: $45 \text{ MHz} \div (2 \times 25 \text{ KHz}) = 900 \text{ channels}$.

2. * Each cell is allocated 80 voice channels.

calculation: $560 \text{ total channels} \div 7 \text{ cell cluster} = 80 \text{ channels per cell}$.

3. * The GSM deployment provides 800 simultaneous voice channels.

calculation: $100 \text{ carriers} \times 8 \text{ time slots} = 800 \text{ channels}$.

4. * The power control ratio is approximately 13 dB.

calculation: $10 \log_{10} (80/4) = 13 \text{ dB}$.

5. * The carrier to interference (C/I) ratio is 32 dB.

calculation: $-60 \text{ dBm} - (-92 \text{ dBm}) = 32 \text{ dB}$.

6. * Each cell receives 20 channels after frequency reuse.

calculation: $(42 \text{ MHz} \div 300 \text{ KHz}) \div 7 = 20 \text{ channels}$.

7. * The received power becomes 4W.

* Doubling the distance with a path-loss exponent of 4 reduces power by 16 times.

8. * The transmit power must be increased by 4 times.

calculation: $P_2 = P_1 \times 2^2 = 4P_1$.

9. * The cluster has 60 voice channels.
calculation: $4 \times 15 = 60$.
10. * The doppler shift is 83 Hz.
calculation: $f_d = \frac{V}{c} \times 900 \times 10^6 = \frac{27.78 \times 900 \times 10^6}{3 \times 10^8} = 83$
11. * Each TDMA slot lasts 0.625 ms.
calculation: $5 \text{ ms} \div 8 = 0.625 \text{ ms}$.
12. * 180 carrier frequencies can be created.
calculation: $24 \text{ MHz} \div 200 \text{ KHz} = 24000 / 133 = 180$.
13. * The free received power is 6 dBm.
calculation: $8 \text{ W} = 39 \text{ dBm}$.
 $39 - 33 = 6 \text{ dBm}$.
14. * The free space path loss is 106 dB.
calculation: $\text{FSL} = 32.44 + 20 \log_{10}(1900) + 20 \log_{10}(2.5) = 106 \text{ dB}$.
15. * The cluster can handle 1500 simultaneous
calculation: $6 \times 250 = 1500$.
16. * The system supports 720,000 active calls
calculation: $4000 \times 180 = 720,000$.

rels.

$$\frac{\times 10^6}{10} = 88 \text{ Hz}$$

ated.

$$4000 / 200$$

1.

B.

$$100) +$$

Simultaneous

cells

17. * The base station uses 3.75 MHz
calculation: $150 \times 25 \text{ KHz} = 3750 \text{ KHz} = 3.75 \text{ MHz}$

18. * The guard bands waste 1.16 MHz.
calculation: $(30 - 1) \times 40 \text{ KHz} = 29 \times 40 = 1160 \text{ KHz}$
 $= 1.16 \text{ MHz}$

19. * Approximately 30 handoffs occur in one hr.
calculation: cell diameter = $2 \times 1.5 = 3 \text{ km}$
distance in 1 hour = 90 km.
Handoffs = $90 / 3 = 30$.

20. * The network can support 80,000 users.
calculation: $8 \times 40 \times 250 = 80,000$.

21. * Approximately 2500 cells are required to cover the area.
calculation: $400 \div (0.4)^2 = 400 \div 0.16 = 2500 \text{ cells}$.

22. * The FDMA system supports 200 channels.
calculation: $6 \text{ MHz} \div 30 \text{ KHz} = 6000 \div 30 = 200$.

23. * The total offered traffic is 200 Erlangs.
calculation: $1200 \times 4 \times \frac{2.5}{60} = 200 \text{ Erlangs}$.

24. * The network can serve 100 users simultaneously
calculation: $40 \text{ MHz} \div 400 \text{ KHz} = 100$.

25. * The total user capacity is 720,000 users.
calculation: $600 \times 1200 = 720,000$.

26. * A maximum of 50 paging rounds are needed.
calculation: $200 \div 4 = 50 \text{ paging rounds}$.

27. * The blocking probability is 16.7%.
Calculation: offered load = $40,000 \times 3 = 120,000$.
 Blocked = 20,000.
28. * 90,000 subscribers are swarming.
Calculation: $600,000 \times 15$.
29. * The switch is operating at 75% utilization.
Calculation: $(9000 \div 12000) \times 100 = 75$.
30. * Each cell receives 60 channels.
Calculation: $(48 \text{ MHz} \div 800 \text{ KHz}) \div 4 = 240 \div 4 = 60$.
31. * The spectrum supports 150 channels.
Calculation: $30 \text{ MHz} \div 200 \text{ KHz} = 150$.
32. * 2475 handoffs were successful.
Calculation: 2500×99 .
33. Calculation: $12 \text{ W} = 40.8 \text{ dBm}$, $40.8 - 45 = -4.2 \text{ dBm}$.
 * The received power is approximately -4.2 dBm.
34. * The cluster supports 6300 users.
Calculation: $7 \times 900 = 6300$.
35. * Each TDMA slot lasts 0.6 ms.
Calculation: $4.8 \text{ ms} \div 8 = 0.6 \text{ ms}$.
36. * The Doppler shift is approximately 250 Hz.
Calculation: $v = 150 \text{ km/h} = 41.67 \text{ m/s}$; $f_d = 250 \text{ Hz}$.
37. * The blocking percentage is 10.7%.
Calculation: offered load = $70,000 \times 4 = 28,000$.
 Blocked = 30,000.

38. * The cluster capacity is 5400 users.
calculation: $9 \times 600 = 5400$.
39. * 240 calls are dropped every second.
calculation: $60,000 \times 0.4 = 240$.
40. * The signal experience 130 dB path loss.
calculation: $43 - (-87) = 130 \text{ dB}$.
41. * The cell covers 375 km^2 .
calculation: $2.5 \times 150 = 375 \text{ km}^2$.
42. * Each cell gets 40 channels.
calculation: Total channels = $48 \text{ MHz} \div 200 \text{ KHz} = 240$
 $= 240 \div 6 = 40$.
43. * 25 paging cycles are needed.
calculation: $150000 \div 6000 = 25$
44. * The call experience 3 handoffs.
calculation: $9 \div 3 = 3$.
45. * The signal-to-noise Ratio is 30 dB.
calculation: $\text{SNR} = -85 - (-115) = 30 \text{ dB}$.
46. * The cell must provide 37.5 Mbps bandwidth.
calculation: $250 \times 150 \text{ Kbps} = 37500 \text{ Kbps} = 37.5 \text{ Mbps}$.
47. * The total traffic generated is 250 Erlangs.
calculation: $1500 \times 4 \times 2.5/60 = 250 \text{ Erlangs}$.
48. * 3 MHz of bandwidth is reserved for control channels.
calculation: $250 \times 12 \text{ KHz} = 3000 \text{ KHz} = 3 \text{ MHz}$.

49. * The network must support 900 Mbps.
Calculation: $1500 \times 600 \text{ Kbps} = 900000 \text{ Kbps}$
 $= 900 \text{ Mbps}$.

50. * Approximately 7 towers are required.
Calculation: Tower area $= \pi R^2 = 3.14 \times 4^2$
 $= 50.24 \text{ km}^2$.

$$350 \div 50.24 \div 7.$$